

Mohammed Yasser Ismael

DATA SCIENCE STUDENT

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About Me

Data science student skilled in Python, SQL, and Power BI, with hands-on experience in data analysis, visualization, and machine learning. Passionate about extracting actionable insights from real-world datasets to support data-driven decision making.

Education

Faculty of Computers & Artificial Intelligence - Cairo University

2023 – 2027

- GPA: Very Good

Skills

Technical Skills:

- **Python:** Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, SciPy, TensorFlow, PyTorch
- **SQL:** PostgreSQL, MySQL, SQL Server
- **Data Analysis & Visualization:** EDA, Data Cleaning, Feature Engineering, Power BI, Excel
- **Machine Learning:** Regression, Classification, Clustering, Anomaly Detection, Model Evaluation
- **Deep Learning:** CNNs, RNNs, Autoencoders, GANs, Transformers, Diffusion Models
- **AI & NLP:** Natural Language Processing (NLP), Generative AI, Hugging Face, LLM Fundamentals
- **AI Frameworks & Concepts:** Introductory RAG (Retrieval-Augmented Generation), LangChain

Soft Skills:

- Time Management
- Team Collaboration
- Critical Thinking
- Adaptability
- Problem Solving

Work Experience

Data Analyst Intern – Elevvo Pathways

Sep 2025 – Oct 2025

- Cleaned and analyzed four datasets using Python & Excel
- Built dashboards and reports to communicate insights

Data Analyst Intern – Uneeq Interns

Oct 2025

- Explored, cleaned and analyzed three different datasets using Python & Excel
- Built dashboards to communicate insights

Projects

Loan Defaulter Prediction System (Graduation Project – Digital Egypt Pioneers Initiative)

- Led a team of five to build an end-to-end loan default prediction system using ~350K records and 150 features from Kaggle datasets.
- Conducted business understanding, data cleaning, EDA, feature selection, and handled class imbalance using SMOTE and oversampling techniques.
- Trained and evaluated LightGBM, Random Forest, and CatBoost models, achieving the best ROC-AUC and accuracy with LightGBM, and deployed the final model via a web application to predict customer default risk.

Energy Consumption Forecasting System

- Developed an end-to-end time-series forecasting system using historical energy consumption data from 8 PJM-supported utility companies (late 1990s–2018).
- Applied full data science lifecycle: business understanding, data preprocessing, feature engineering (time-based & lag features), model training, validation, and deployment.
- Trained and optimized an XGBoost Regressor model achieving near-perfect fit ($R^2 \approx 0.999$) with minimal prediction error on unseen temporal data.
- Implemented recursive forecasting logic to generate future energy demand predictions beyond 2018.
- Deployed the model via a Streamlit web application enabling dynamic user-defined forecasting and test-data simulation.

Supply Chain Latency & Risk Prediction

- Built an end-to-end ML pipeline on 180K+ orders to classify shipment outcomes into 4 classes (Advance, On Time, Late, Canceled)
- Applied unsupervised learning (PCA, UMAP, KMeans) for exploratory analysis, confirming feature separability before modeling
- Handled severe class imbalance (54% Late Delivery) using a combined SMOTE + RandomUnderSampler strategy
- Tuned XGBoost and LightGBM via RandomizedSearchCV; final XGBoost model achieved 73.7% accuracy and 0.85 F1 on Late Delivery
- Deployed as a Streamlit web app with real-time predictions from raw order inputs

Business Insights & Executive Report for Olist (Elevvo Pathways Internship Project)

- Analyzed Olist's multi-table e-commerce dataset covering customers, products, sellers, and orders to assess overall business performance.
- Produced an executive-level report with insights on customer behavior, product performance, and seller efficiency, and provided data-driven recommendations to leverage Olist's strengths and improve sales outcomes.

Flights Delay & Cancellation Analysis

- Analyzed a large-scale U.S. flight dataset (~5.5M records, 2015) to investigate delay and cancellation patterns.
- Performed data cleaning, exploratory data analysis, and visualization to identify key causes of disruptions, seasonal trends, and optimal travel periods, delivering actionable operational insights.

Courses

Data Scientist (Round 3) - Digital Egypt Pioneers Initiative

IBM AI Engineering Professional Certificate - IBM

Machine Learning Specialization – DeepLearning.AI

Data Scientist in Python – Data camp

Data Analyst in SQL – Data camp

Data Analyst in PowerBI – Data camp